Connor Haley

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Education

University of Florida — Gainesville, FL
Bachelor of Science in Computer Engineering, Sociology Minor, Engineering Innovation Cert. — *Aug. 2020 – Dec. 2025*

Experience

Embedded Systems Lead - DroneHome Design Team — Aug. 2024 – May 2025
University of Florida — *Gainesville, FL*

- Developed vehicle that autonomously locates, retrieves, and recharges field drones to minimize mission downtime.
- Engineered Flask UI to interface with remote R/C vehicle over LTE using ROS2 on NVIDIA Nano Jetson Super
- Created embedded control system integrating sensors, microcontrollers, and comms for real-time vehicle operation
- Implemented modular control software supporting feedback loops, sensor fusion, and autonomous fault recovery.

Engineering Team Lead - Theme Park Engineering & Design (TPED) — Aug. 2020 – May 2023
University of Florida — *Gainesville, FL*

- Led multidisciplinary team creating attraction prototypes integrating mech, electrical, and software subsystems.
- Designed 3D-printable roller coaster in Solidworks and engineered prototype animatronic eyes with Raspberry Pi.
- Oversaw research, schematics, & design to ensure reliable and safe show system operation within ASTM standards

Additional Experience

Technical Lead - UF Cultural Organizations — Aug. 2021 – March 2026
University of Florida — *Gainesville, FL*

- Engineered shows for hundreds of dancers, taking teams to LA, Chicago, North Carolina, & beyond to compete!
- Co-founded two dance teams to compete nationally, engineering audio, physical set, electronic props, & SFX.
- Integrated creative workflows with technical design, applying signal-processing and systems-integration principles.

Projects

Start Ups: RevIt Lighting & RoyalMaccro — Oct. 2024 – Present

- RevIt Lighting is a small team start-up creating LED lights that respond to a car's internal data accessed via OBD-II port. Using ESP32 devices leveraging ESPNOW & Bluetooth to sync car diagnostics and data, such as RPM, to LEDs via app. Working with engineers from IC2, a Gainesville embedded device engineering company.
- RoyalMaccro is a personal start-up; developed an online multiplayer global intelligence sim with 220+ countries.

Embedded Systems & Real-Time Communication Labs — Aug. 2025 – Dec. 2025

- Configured UART, I^2C, and SPI interfaces on a TM4C123 µC to communicate with sensors & LCD displays.
- Developed multi-threaded RTOS apps with priority scheduling, IPC, dynamic threading, & blocking semaphores.
- Configured Tiva DSP, Beaglebone Connection, Audio Output, & my personal real-time light show app, StarTOS.

Technical Skills & Engineering Tools

Programming Languages: C, C++, Python, Java, Assembly (ARM), Verilog, VHDL, MATLAB, SQL (PostgreSQL)
Hardware & Embedded Systems: Raspberry Pi, NVIDIA Jetson Nano, FPGA Design, µCs, µPs, UART, I^2C, SPI
Engineering Tools & Design: ROS2, FastAPI, Socket.io, Atmel Studio, LTSpice, Quartus Prime, SolidWorks, KiCad
Development & Integration: Git, Visual Studio, Full Stack Web Development, Docker, WebSockets, API Integration

Relevant Coursework

Microprocessors II (Real-Time Operating Systems (RTOS)), Microprocessors I, Digital Design & FPGA Implementation, Digital Logic, Computer Engineering Design I & II, Data Structures & Algorithms, Computer Organization, Engineering Leadership, Operating Systems, Intro to Digital Signal Processing (DSP), Intro to Software Engineering

Leadership & Interests

Passionate about embedded software, real-time hardware integration, and control systems, complemented by experience in full-stack and interactive web design. Background in customer-facing roles (banking, retail, security) reinforced communication, reliability, and composure under pressure. Dedicated to leading teams through complex challenges.